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SUPERSEDING
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MILITARY SPECIFICATION

SEALING MACHINES, ELECTRICAL IMPULSE

(JAW TYPE)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers portable and bench mounting electrical impulse, jaw type sealing machines for thermoplastic films.

1.2 Classification.- Sealing machines shall be of the following classes, types, and sizes as specified (see 6.2):

Class 1 - Machines with an electrical impulse element in one jaw
Class 2 - Machines with electrical impulse elements in both jaws

Type I - Portable, Sealing Machine, mechanical operation, separate control unit

Size 6 - 6" jaw length
Size 8 - 8" jaw length
Size 10 - 10" jaw length

Type II - Bench Mounting, Sealing Machine, compressed air operation, integral control unit

Size 12 - 12" jaw length
Size 14 - 14" jaw length
Size 16 - 16" jaw length
Size 18 - 18" jaw length
Size 20 - 20" jaw length
Size 24 - 24" jaw length
Size 28 - 28" jaw length
Size 36 - 36" jaw length
Size 48 - 48" jaw length

FSC 3540

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Type III - Bench Mounting, Sealing Machine, Hand or Foot Operation, integral control unit

Size 12 - 12" jaw length
Size 14 - 14" jaw length
Size 16 - 16" jaw length
Size 18 - 18" jaw length
Size 20 - 20" jaw length
Size 24 - 24" jaw length
Size 36 - 36" jaw length
Size 48 - 48" jaw length

Type IV - Portable Sealing Machine, Magnetic Operation, separate control unit

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	-	Plating, Cadmium (Electrodeposited)
ZZ-H-496	-	Hose, Pneumatic, Braided
PPP-B-601	-	Box, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-C-4109	-	Coupling Assembly, Air-Hose, Quick-Operating

STANDARDS

FEDERAL

FED-STD-101	-	Preservation, Packaging and Packing Materials: Test Procedures

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc. (UL) Standards

(Application for copies of UL publications should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., N.Y. 11746; 207 East Ohio Street, Chicago, Ill. 60611; or 1655 Scott Boulevard, Santa Clara, California 95050.)

National Fire Protection Association (NFPA)

Standard No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- When specified, the supplier shall furnish a sample for first article inspection and approval (see 4.2, 6.2 and 6.3).

* 3.2 Standard product.- Sealing machines furnished under this specification shall be the manufacturer's current standard product except for those changes required for compliance with the requirements of this specification.

* 3.3 Material.- Materials not definitely specified shall be of the quality normally used in the manufacture of sealing machines, provided the completed item complies with all the provisions of this specification.

* 3.4 Codes and standards.-

* 3.4.1 UL.- The wiring, fittings and conduit, as applicable, shall conform to the appropriate Underwriter's Laboratory (UL) standards for use in non-hazardous locations.

* 3.4.2 NFPA.- The sealing machines shall be wired to conform to NFPA Standard No. 70.

* 3.4.3 Certification.- Prior to approval of the first article or if none is submitted, prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative, that the sealing machines he proposes to furnish meet the requirements specified in 3.4.1 and 3.4.2.

* 3.4.3.1 UL.- Acceptable evidence of meeting the requirements of 3.4.1 shall be the UL label or listing mark, or a certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the wiring, fitting and conduit, as applicable, conform to the appropriate UL Standards.

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* 3.4.3.2 NFPA.- Acceptable evidence of meeting the requirements of 3.4.2 shall be the manufacturer's certified statement that the sealing machine is in accordance with NFPA Standard No. 70.

* 3.5 Design and construction.- All sealing machines shall be of the jaw type, equipped with electric impulse elements (see 6.4.1), providing not less than 250 watt capacity per square inch of element and uniformly heating the entire sealing area (see 4.4.4). Electrical components shall be designed for the voltage frequency and number of phases specified (see 6.2). Operational and functional testing shall be performed at the specified voltage, frequency and number of phases (see 4.4 and 6.2). A red pilot light will indicate when the master switch is closed and an amber pilot light will indicate when a sealing cycle is in progress. Machines using compressed air shall be equipped with separator - strainers and lubricators in accordance with commercial practice for clean-dry air and lubrication of cylinders and associated operating components. Design features common to all machines shall include provision for re-lubrication of friction points, easy replacement of wear parts, free operation of jaws at all temperatures, protection of the operator from contact with parts at elevated temperatures, and both jaws shall have replaceable covers of polytetrafluorethylene glass cloth or equal to prevent adhesion of the work to the jaws. Cords connecting foot switch to units shall be UL approved and not less than 6 feet long.

* 3.5.1 Sealing cycle.- The operation of sealing cycle shall be such that both heating and cooling cannot be interrupted nor the heating or cooling times changed after the operating switch has been activated. Reactivation of the operating switch during any cycle shall have no effect on the heating or cooling cycle, and cannot be repeated until operating switch has been deactivated. An amber pilot light which is wired in the circuit shall operate during the sealing cycle, (heating and cooling) (see 4.4.1).

* 3.5.2 Master switch.- A master switch, readily accessible to the operator, shall be provided on all sealing machines to energize the control unit and provide emergency termination of the sealing cycle.

* 3.5.3 Indicator lamps.- A red indicator lamp (pilot light) shall indicate that the controls are energized and that the sealing machine is ready to operate. An amber indicator lamp (pilot light) shall indicate that a sealing cycle is in progress.

* 3.5.4 Timers.- All sealing machines shall have timers to control the heating and cooling periods. The heating timer shall have an infinitely variable range from 0 to 2 seconds minimum. The cooling timer shall have an infinitely variable (stepless) range from 0 to 6 seconds minimum. Provisions

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shall be made for accurate control of both the heating and cooling time cycles. Both infinitely variable timers shall have scale calibration intervals of 1/10 to 1/2 second. (see 4.4.5).

* 3.5.5 Fuses.- Electrical heating elements shall be protected by a separate circuit switch with a fuse or manual reset circuit breaker.

* 3.5.6 Spring tension.- Each electrical impulse heating element shall be held firmly and securely in place by means of springs under tension. Impulse elements shall evidence no kinking after being operated for one hour (see 4.4.6).

* 3.5.7 Jaws.- At least one of the jaws shall have a resilient silicone (or equal) rubber cushion to compensate for variation in thickness and wrinkled lap joints. If heating elements are in both jaws (class 2 machines), the elements shall be so placed in the resilient jaw cushion that the effective heat of the element is not reduced. Both jaws shall be provided with readily removable anti-stick covers made of polytetrafluorethylene coated fiberglass material or equal that may be easily repaired or renewed if damaged or worn.

* 3.5.8 Electric cable.- Unless otherwise specified (see 6.2), sealing machines shall be provided with a 6-foot minimum length flexible cord and connector meeting the requirements of the National Electrical Code and Underwriters' Laboratories (see 3.5). Flexible cords connected to foot switches shall be UL approved and not less than 6 feet long.

* 3.5.9 Dielectric strength.- Electrical insulation of current carrying components of the sealing machines shall have a dielectric strength meeting the requirements of the National Electrical Code and Underwriters' Laboratories (see 4.4.3).

3.5.10 Electromagnetic compatibility.- When specified (see 6.2), the sealing machines shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.7).

* 3.5.11 Type I, portable machines.- Each sealing machine shall consist of a sealing unit with a separate control unit. The sealing machine and control unit shall be interconnected by means of a ten foot minimum length U.L. approved electric cable (see 6.2). The control unit shall contain a heating timer, cooling timer, main switch, pilot light, and electrical components and connections essential to the performance of the sealer. This type sealer is operated by pressing a switch mounted on the sealer convenient to finger or thumb touch. When activated, the switch shall close an electrical circuit to start the sealing cycle and light the amber pilot light when predetermined pressure has been applied. Predetermined jaw pressure shall be maintained to provide a good and distortion free seal (see 4.4.1).

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3.5.11.1 Handle.- The type I portable heat sealing machine shall have the sealing unit handle and operating switch insulated so that when the sealer is continually operated as specified in 3.5.6 for one hour, the temperature of the handle and operating switch shall not exceed 100°F. (see 4.4.6).

3.5.11.2 Weight.- The weight of type I sealing machines, less cable and remote control cabinet, shall be not more than 2 pounds, 10 ounces for size 6, 2 pounds, 14 ounces for size 8, and 3 pounds, 2 ounce for size 10.

* 3.5.11.3 Remote control unit.- Type I sealing machine remote control unit shall contain all necessary parts and accessories and comply with applicable requirements of the National Electrical Code and Underwriters' Laboratories.

* 3.5.12 Type II, bench mounting sealing machines (compressed air).- Movable jaws of type II machines shall be activated through mechanical linkage connected to air operated cylinders. The machine shall be a complete unit with integral controls and shall be equipped with an air gage, separator-strainer and lubricator, and a regulator for the control of the jaw pressure. Each electrical impulse element shall have separate current overload protection. The sealing machine shall have a separate operating switch interconnected to the sealer with a ten foot minimum length cable. (see 6.2). The sealing unit shall be rigidly mounted on a metal base. Mounting holes shall be provided for securely bench-mounting the sealing unit base. The design of the machine shall be such that a continuous web of material can pass through the machine. The web shall be no wider than 6 inches less than the maximum sealing length of the machine.

3.5.12.1 Sealing cycle.- The type II sealer shall be operated by means of a hand or foot switch, which when activated, will close an electrical circuit starting the sealing cycle and lighting the amber pilot light. Predetermined jaw pressure shall be reached prior to starting the heat sealing cycle. This cycle shall include the transmittal of air pressure to the sealing jaws, followed by the heating time cycle and the cooling time cycle for providing distortion free seals. The jaws shall be maintained in the sealing position during the entire operation. At the completion of the cycles, the jaws shall open automatically. The entire cycle shall be automatic even though the hand or foot switch has been released prior to the completion of the sealing cycle. The cycle shall not repeat until the switch has been released and reactivated. The machine should be so constructed that the entire sequence has to be completed before any new sequence can start (see 4.4.1).

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3.5.12.2 Effective sealing pressure gage.- Type II sealing machines shall have a pressure gage to indicate the average effective sealing pressure. The gage shall be calibrated in steps of 2 pounds per square inch (psi) of actual applied pressure (see 4.4.6). The gage shall indicate the average effective sealing pressure over the area being sealed to within plus or minus 10 percent. The words effective pressure shall be permanently and legibly marked on the face of the gage. A reference scale corresponding to the dial reading may be used in place of the pressure gage.

3.5.12.3 Effective sealing pressure controller.- Type II sealing machines shall have means to make rapid adjustments for the effective, calibrated sealing pressures throughout the range from 5 to 40 psi (see 4.4.1).

3.5.12.4 Integral control unit.- The integral control unit of the type II machine shall be located on the sealing unit in a position convenient to the operator, and shall contain a master switch, indicator lamp, pilot lamp, effective sealing pressure gage, effective sealing pressure controller, calibrated dwell and cooling timers, and all parts and accessories necessary for control of the sealing operation. All applicable components shall be securely attached and suitably located.

3.5.12.5 Air hose.- When specified (see 6.2), type II sealing machines shall be equipped with pneumatic hose conforming to ZZ-H-496 for connection to a compressed air source. The free end of the air hose shall terminate in a quick-operating air hose coupling assembly conforming to MIL-C-4109.

3.5.12.6 Air pressure regulator.- Type II sealing machines shall have an air pressure regulator, controlled by means of an external knob, capable of regulating 100 psi maximum inlet air down to the air pressure required to produce effective sealing pressures from 5 to 40 psi (see 4.4.1).

3.5.12.7 Air valve.- Type II sealing machines shall have a solenoid-operated valve for supplying air to the sealing unit (see 4.4.1).

3.5.13 Type III, bench mounting sealing machines, hand or foot operation.- The design of the type III machine shall provide for bench mounting and application of pressure by hand or by foot pedal. The machine will be a complete unit with integral controls including heating timer, cooling timer, main switch, and electrical components essential to the operation of the sealer. The heating element shall have current overload protection. The design of the machine shall be such that a continuous web of transparent material can pass through the machine. The web shall be no wider than 6 inches less than the maximum sealing length of the machine. This type sealer shall be actuated by pressing a hand lever or foot pedal. When a predetermined jaw pressure is attained by actuation of the lever or pedal, the sealing cycle is initiated

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and continues until the operator releases the pressure. An amber pilot light shall indicate the duration of the sealing cycle. Jaw pressure shall be provided by tension springs or other means and will hold the material tightly. At the end of the sealing cycle the amber light will go out indicating that the operator should release the pressure on the lever or pedal. All the controls will then automatically reset, ready for the next sealing cycle (see 4.4.1).

3.5.13.1 Integral control unit.- The integral control unit shall be located on the type III sealing unit in a position convenient to the operator. The integral control panel shall contain a master switch, indicator lamp, pilot lamp, calibrated heating and cooling timers, and all parts and accessories necessary for control of the sealing operation. All applicable components shall be securely attached and suitably located.

* 3.5.14 Type IV, portable machines, (magnetic operation).- The sealing machine shall be similar to the type I in design and construction except that jaw closure and controls shall be electromagnetically powered.

* 3.6 Interchangeability.- Every part with a common part number shall be functionally and dimensionally interchangeable.

* 3.7 Performance.- Sealing machines shall be capable of sealing a minimum of 24 mils of low density polyethylene layered material within two seconds. Sealing machines shall form seals exhibiting no separation or delamination when tested in accordance with method 2024 and method 5009 of FED-STD-101 (see 4.4.2.1, 4.4.2.2 and 6.2). Homogenous seals shall be formed in the presence of wrinkles, lap joints, overlaying lap joints and contamination of the heat sealable interface. Jaws of the sealing machines shall not stick to the thermoplastic upon completion of the sealing cycle.

3.8 Finishes and protective coatings.-

* 3.8.1 Painting.- All external surfaces, except corrosion resistant steel, anodized aluminum, working surfaces, threads, and cadmium plated parts, shall be finished in accordance with the manufacturer's standard practice.

3.8.2 Plating.- All metal surfaces except stainless steel, aluminum alloy, or where painting is not applicable, shall be cadmium plated in accordance with type I, class b, of QQ-P-416.

* 3.8.3 Anodizing.- Where applicable, all aluminum alloy parts shall be anodized in accordance with the manufacturer's standard practice.

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3.9 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The name plate shall include the following:

Sealing Machine, Electrical Impulse (Jaw Type)
Military Specification MIL-S-22783C
Type _____ *
Class _____ *
Size _____ *
Manufacturer's Model Number
Manufacturer's Name

*Applicable data shall be entered by the contractor.

* 3.10 Workmanship.- Sealing machines shall be free of blemishes, slivers, splinters, burrs and sharp edges. Soldering, welding and brazing shall be sound and thorough and shall be free of flux, porosity, slag, corrosion and projections. All fastening devices shall be drawn tight, electrical wiring shall be secure at terminals with adequate slack to prevent strain and permit re-lugging. Coated, plated or treated finished surfaces shall be smooth, adherent and free of areas of no film, or imbedded foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a first article is required, it shall be examined for defects listed in table I, dimensions specified, and tested as specified in 4.4.1 through 4.4.6 and 4.4.7, if applicable. The presence of any visual defect, any dimension not within specified requirements, or failure to pass any test, shall be cause for rejection of the preproduction sample.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 Process examination.- Examination shall be made to insure that primer is applied, when applicable, to all external surfaces except corrosion resistant steel, anodized aluminum, working surfaces, threads and cadmium plated parts. When nonconformance is noted, correction shall be made to the item and to the process.

* 4.3.3 End product inspection.- A lot shall consist of all complete sealing machines of one class, type and size offered for inspection at one time. The sample unit shall be one complete sealing machine.

* 4.3.3.1 Visual examination.- The completely assembled end item shall be examined for defects listed in table I. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified		X
	Not smooth and adherent	X	
	Not free of areas of no finish and embedded foreign matter		X
Workmanship and general construction	Blemishes, burrs, slivers, splinters and sharp edges	X	
	Parts broken or punctured	X	
	Parts missing or malformed	X	
	Parts loose		X
	Any component not as specified	X	
	Any component not in proper alignment	X	
	Any component out of adjustment	X	
Welding, brazing, soldering	Not sound and adherent	X	
	Not free of flux, porosity, corrosion and projection		X
Fasteners	Missing, broken, stripped or loose	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Electrical components	Insulation cut, abraded	X	
	Wiring not properly joined or loose at terminals	X	
Marking identification	Missing, incomplete, not legible, or not as specified	X	

* 4.3.3.2 Dimensional and weight examination.- Examination shall be made of the sealing machines to determine compliance with dimensional and weight requirements. Any weight or dimension that exceeds the specified requirement or tolerance shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 defects expressed in defects per hundred units.

* 4.3.4 Code and standard compliance.- Proof of compliance with requirements of 3.4 shall be available to the Government representative.

* 4.3.5 Testing of the end item.- When a first article is not required, the initial production specified unit shall be tested as specified in 4.4.1 through 4.4.6 and 4.4.7, if applicable. Failure to pass any test shall be cause for rejection of the initial unit. The specified sealing machines offered for examination shall be tested as specified in 4.4.1 through 4.4.6. The inspection level shall be II with an AQL of 4.0 defects expressed in defects per hundred units.

4.3.6 Examination of preparation for delivery.- Examination will be made to determine that preservation, packing and marking as required by section 5 is complied with. Defects will be scored as specified below. The inspection level shall be S-4 with an acceptable quality level of 6.5 defects per hundred units.

Level of preservation and packaging and packing not as specified (see 5.1 and 5.2).
Method of preservation, when required, improperly applied or missing.
Boxes not class or style specified.
Boxes improperly closed; strapping omitted.
Cushioning, blocking and bracing, as required, inadequately applied or missing.
Markings on interior and exterior containers omitted, incorrect or illegible.

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* 4.4 Tests.- Operational and functional tests shall be performed at the voltage, frequency and number of phases specified (see 3.5 and 6.2).

* 4.4.1 Operational test.- Sealing machines shall be connected to an electrical power supply and air supply when applicable, and operationally tested. The following items shall be checked, as applicable:

a. Uninterrupted heating and cooling cycle (see 3.5.11, 3.5.12 and 3.5.13).
b. Remote control unit governs the heating timer, cooling timer, and pilot lights (see 3.5.11).
c. Switch when activated starts sealing cycle (see 3.5.11 and 3.5.12).
d. Pilot light on for duration of sealing cycle (heating and cooling) (see 3.5.1, 3.5.11, 3.5.12 and 3.5.13).
e. Jaw pressure is maintained by springs or other means (see 3.5.11, and 3.5.13).
f. Jaw pressure maintained by air (see 3.5.12).
g. Air regulator with gage controls seals jaw pressure (see 3.5.12 and 3.5.12.6).
h. Jaw pressure applied before sealing cycle starts (see 3.5.12).
i. Jaws open automatically after completion of sealing cycle (see 3.5.12 and 3.5.13).
j. Type III machines energized by pressing handle or lever (see 3.5.13).
k. Pressure applying means is locked when sealing cycle is initiated (see 3.5.13).

Failure to conform to any of the above tests shall constitute a defect.

* 4.4.2 Performance.- Each type and class of sealing machines shall be tested as follows to determine compliance with 3.5 through 3.5.13.1, 3.7 and 6.2.

* 4.4.2.1 Sealing test.- Unless otherwise specified (see 6.2) low density polyethylene shall be used for this test. The test shall be conducted by sealing a total thickness of not less than 24 mils of low density polyethylene using a two second heat cycle. Any indication of failure to make a seal as specified in 3.7 and method 2024 of FED-STD-101 shall constitute failure of this test.

* 4.4.2.2 Seam leak test.- Three 6 x 6 inch pouches shall be fabricated from six pieces of 4 mil low density polyethylene. At least one of the pouches shall have a wrinkle extending the entire width of a seam, either in the bottom seam or within two inches of it. A water solution (see Method 5009 of FED-STD-101) containing 1 per cent Aerosol OT, or equivalent, with sufficient

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dye to produce a distinct color shall be poured into each pouch to a level of 2 inches above the top of the bottom seam. The three assemblies shall be suspended for one hour at room conditions after which the dye solution shall be removed and the pouches examined. Any evidence of leakage at the seams shall be cause for rejection. When specified, (see 6.2), the seam seal shall additionally be tested in accordance with the vacuum chamber technique of Method 5009 of FED-STD-101. Any evidence of leakage shall constitute failure of this test.

* 4.4.3 Dielectric test.- Insulation shall meet the dielectric requirements of the National Electric Code and the Underwriters' Laboratories when tested for use at the voltage, frequency and number of phases specified (see 3.5, 3.5.9 and 6.2). Failure to withstand the applied voltage for not less than one minute shall constitute failure of this test.

* 4.4.4 Heating element test.- The sealing machine time control shall be turned to the maximum dwell time indicated on the timer. Sealing surface wattage shall be determined by thermocouple array, pyrometer, or other standard means. Measurements shall be taken at not less than three points (midpoint and each end). Any result indicating less than 250 watts shall constitute failure of this test.

* 4.4.5 Dwell time and cooling time.- The time to complete the heating and cooling cycles shall be verified as an infinitely variable (stepless) range of time from 0 to 2 seconds minimum for the heating cycle and 0 to 6 seconds minimum for the cooling cycle. Failure to meet these cycle requirements as specified in 3.5.4 shall constitute failure of this test.

* 4.4.6 Temperature, distortion and effective pressure test.- All machines shall be operated continuously for one hour. Types I, II, and IV with 2 second heating, 2 second cooling, and 2 second loading time cycles and type III with 2 second heating, 6 second cooling, and 6 second loading time cycles, using material as specified in 4.4.2.1. Any evidence during or upon completion of the test, of kinking of heating elements (see 3.5.6), heating of type I handles and switches in excess of 100°F (see 3.5.11.1), or deviation exceeding 15% in calibration accuracy of effective sealing pressure gages on type II machines (see 3.5.12.2) shall constitute failure of this test.

4.4.7 Electromagnetic compatibility.- When electromagnetic compatibility is required, the first article or initial unit of production, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461.

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* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Disassembly.- Each sealing machine shall be disassembled, as necessary, to reduce cube and prevent damage while in transit. Disassembly shall be such as not to require special skills or tools during assembly. Disassembled parts shall be match marked.

5.1.2 Level A.-

5.1.2.1 Cleaning and drying.- The sealing machine shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1 and D-4 of MIL-P-116.

5.1.2.2 Unit package.- The sealing machine shall be packaged in accordance with method IIa or IIb of MIL-P-116. Component parts and accessories removed through disassembly shall be cushioned and placed in a close fitting fiberboard box conforming to PPP-B-636, Class - Domestic. Boxes shall be closed in accordance with the appendix of PPP-B-636. The container shall be secured within the method II package to prevent movement that may rupture the heat sealed barrier material. The outer container of the unit package may be the shipping container specified for level A or B packing (see 5.2.1 and 5.2.2), and additional packing need not be provided.

5.1.3 Level C.- The sealing machine shall be cleaned, preserved with operational lubricants and unit packaged in a fiberboard carton conforming to PPP-B-636 together with tools, assembly instructions and operation instructions. The unit container shall meet the requirements for level C packing (see 5.2.3). Tools, component parts and instructions shall be individually wrapped or packaged within the unit container.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- Each complete sealing machine, preserved and packaged as specified in 5.1, shall be packed in a cleated plywood or a fiberboard shipping container conforming to overseas type PPP-B-601, or grade V2s, style RSC of PPP-B-636. Blocking, bracing and cushioning shall conform to MIL-STD-1186. Closure and strapping shall be in accordance with the appendix of the applicable container specification. Weight of contents of each fiberboard shipping container shall not exceed 90 pounds.

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5.2.2 Level B.- Each complete sealing machine, preserved and packaged as specified in 5.1, shall be packed in a cleated plywood or a fiberboard shipping container conforming to domestic type of PPP-B-601, class 1, style E, F or G of PPP-B-640, or type CF (variety SW) or SF, class domestic, grade 275, style RSC of PPP-B-636. Blocking, bracing and cushioning shall conform to MIL-STD-1186. Closure shall be in accordance with the appendix of the applicable container specification. Weight of contents of each fiberboard shipping container conforming to PPP-B-636 shall not exceed 65 pounds.

5.2.3 Level C.- Sealing machines shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The sealing machines covered by this specification are intended for use in impulse sealing of thermoplastic materials for preservation and storage of supply and maintenance components.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and size required (see 1.2).
(c) Electrical characteristics required (voltage, phase and frequency (see 3.5).
(d) When connection cable more than 6 feet in length is required (see 3.5.11 and 3.5.12).
(e) When finish or color is to be other than as specified (see 3.8.1).
(f) Power supply cable if other than specified (see 3.5.8).
(g) When and what length pneumatic hose and coupling is required (see 3.5.12.5).
(h) Plastic material for test, if other than specified (see 4.4.2.1).
(i) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(j) When a first article is required (see 3.1, 4.2 and 6.3).
(k) When electromagnetic compatibility is required (see 3.5.10).
(l) When vacuum chamber technique of Method 5009 of FED-STD-101 is to be followed (see 4.4.2.2).

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6.2.1 Contract data requirements.- Any requirements for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.3 First article.- When a first article is required, it should be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one complete sealing machine. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.4 Definition.- For the purpose of this specification, the following definition will apply.

6.4.1 Electrical impulse heating element.- The element is made from a material with a low thermal inertia which may be heated by means of electric current, thus bringing the temperature of the jaw above the melting or softening point of the thermoplastic. When no current flows, the jaw and element will conduct residual heat away from the thermoplastic material.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - SA
Air Force - 84

Preparing activity:

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